blood

721.CLINICAL ALLOGENEIC TRANSPLANTATION: CONDITIONING REGIMENS, ENGRAFTMENT, AND ACUTE TRANSPLANT TOXICITIES | NOVEMBER 13, 2019

Clostridium Difficile Infection Among Hematopoietic Stem Cell Transplant Recipients and Subsequent Development of Graft-Versus-Host Disease

Shanee Chung, MD,[1] Yasser Abou Mourad, MD[1]

[1]The Leukemia/Bone Marrow Transplant Program of BC, British Columbia Cancer Agency, Vancouver, Canada

bloodjournal Blood blood **(2019) 134 (Supplement_1) : 4494.**

http://doi.org/10.1182/blood-2019-129663

Introduction:

Clostridium difficile (C. difficile) is an anaerobic gram positive rod that colonizes 20-40 % of hospitalized patients and the most common cause of antibiotic-associated diarrhea. The clinical manifestations of C. difficile infection (CDI) range from asymptomatic carriage to severe fulminant infection, leading to bowel perforation and death. Stem cell transplantation (SCT) recipients are at risk of CDI due to prolonged neutropenia, use of broad-spectrum antibiotics, mucosal damage and changes in intestinal microbiota induced by cytotoxic chemotherapy, microbial co-infection and possibly graft-versus-host disease (GVHD). The incidence of CDI amongst patients undergoing autologous SCT varies between five and 15 % and among those undergoing allogenic SCT 12 and 34% in literature[1-5]. Some studies have found a strong association between CDI and GVHD, gut GVHD and/or severe GVHD[1-4] but this was not universally demonstrated[5]. Many studies looking at CDI in the SCT setting are limited by the ubiquity of traditional risk factors for CDI and small sample sizes.

Methods:

In this retrospective observational study, we aimed to review the epidemiology and clinical management of CDI in our autologous and allogeneic SCT recipients and subsequent clinical outcomes including development of GVHD. All recipients of autologous and allogeneic SCT at Vancouver General Hospital over the 10-year period between January 1, 2008 and December 31, 2017, who had diarrhoea and at least one positive stool test for C. difficile toxin by polymerase chain reaction were eligible for inclusion. Patient demographics, transplant details including GVHD prophylaxis, treatment for CDI, complications from CDI and subsequent development of acute and chronic GVHD and survival outcomes were collected by means of review of their electronic and written clinical records. An ethics approval was obtained from the hospital ethics committee.

Results:

Out of the 2104 patients who underwent SCT during the study period, 277 (13.2%) had at least one episode of C. difficile positive diarrhea. The incidence was higher among recipients of allogeneic SCT (19.6%) compared to autologous SCT recipients (9.6%). The majority of the patients (86.7%) were treated with oral metronidazole as a first line therapy and 18.9% required a second line treatment, with oral vancomycin in most cases, due to persistent symptoms with toxin positivity. 21% had recurrence of CDI after an initial successful treatment and 9.4% had two or more recurrences. There was no death directly attributable to acute CDI and the rate of major complications was low. Two patients required an intensive care unit admission with fulminant CDI and one patient underwent bowel resection. 57% of allogeneic SCT patients with post-transplant CDI subsequently developed acute GVHD (45.8% grade 2 or higher acute GVHD, 34% gut GVHD). 60.6% developed chronic GVHD (46.9% moderate to severe GVHD, 19.2% gut chronic GVHD). The median progression free survival was 670.5 days (range 1-3938) and the overall survival time was 984.5 days (range 10-3938). The commonest cause of death was malignancy relapse.

Conclusion:

The incidence of CDI among autologous and allogeneic SCT recipients in British Columbia was 9.6% and 19.6% respectively. There was no death immediately attributable to CDI and few patients developed fulminant CDI. Of the SCT patients who had CDI, 57% and 60.6% subsequently developed acute and chronic GVHD respectively.

References: Trifilio SM et al . Changing epidemiology of Clostridium difficile-associated disease during stem cell transplantation. Biol Blood Marrow Transplant. 2013;19: 405-409.Chakrabarti et al.Clostridium difficile infection in allogeneic stem cell recipients is associated with severe graft-versus-host disease and non-relapse mortality. Bone Marrow Transplant. 2000; 26(8):871-6.Alonso CD et al. Epidemiology and outcomes of Clostridium difficile infections in hematopoietic stem cell transplant recipients. Clinical Infectious Diseases. 2012;54(8):1053-63.Dubberke ER et al. Clostridium difficile-associated disease in allogeneic hematopoietic stem cell transplant recipients: risk associations, protective associations, and outcomes. Clin Transplant. 2010;24(2):192-198.Kinnebrew MA et al. Early Clostridium difficile infection during allogeneic hematopoietic stem cell transplantation. Plos ONE 2014; 9(3):e90158.

Disclosures

No relevant conflicts of interest to declare.

Author notes

*Asterisk with author names denotes non-ASH members.